Homework 2

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1 PSOLA pitch and duration modification

1.1 Apply PSOLA

In this section, we use PSOLA to generate five modified versions of each audio clip:

- one with the pitch raised by a factor of 1.5
- one with it lowered by a factor of 0.75
- one made 0.8 times as long (faster)
- one made 1.5 times as long (slower)
- one in which the pitch contour is replaced by a constant equal to the clip's median F0, which turns the speaker into a monotone

These modifications are applied to both *clean_lj* and *clean_jfk*. Table 1 shows the duration and F0 ratios, along with the F0 standard deviations, for each modified audio clip. When using PSOLA to modify pitch, the estimated F0 ratios closely match the target values, while the duration ratios remain unchanged. Similarly, when modifying duration, the measured duration ratios closely match the target values, while the original F0 is preserved. For the monotone modification, where the pitch contour is replaced with a constant value equal to the median F0, the estimated F0 standard deviation is substantially reduced.

Table 1: Measured duration and F0 ratios for PSOLA-modified audio clips. F0 standard deviations before and after modification are reported for the monotone versions.

Clip	Modification	Duration Ratio	F0 Ratio	F0 Std. Before	F0 Std. After
clean_lj	Pitch ×1.5	1.000	1.495	–	–
	Pitch ×0.75	1.000	0.752	–	–
	Duration ×0.8	0.800	1.003	–	–
	Duration ×1.5	1.500	1.002	–	–
	Monotone	1.000	1.000	67.631	0.697
clean_jfk	Pitch ×1.5	1.000	1.496	–	–
	Pitch ×0.75	1.000	0.775	–	–
	Duration ×0.8	0.800	1.005	–	–
	Duration ×1.5	1.500	1.002	–	–
	Monotone	1.000	1.000	39.576	8.453

We also plot the estimated F0 contours of the original *clean_lj* audio and its pitch-modified versions in Figure 1. The modified F0 contours generally follow the same pattern as the original, but their values are scaled according to the target pitch factors

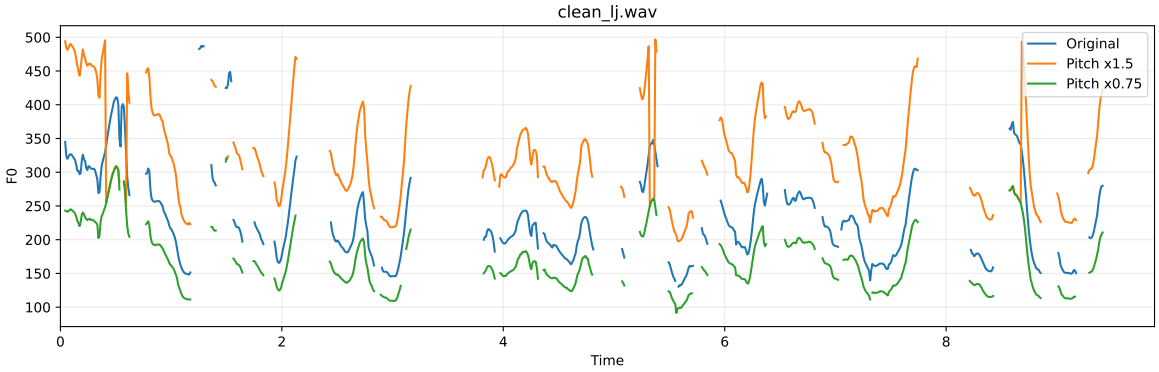


Figure 1: F0 contours of the original *clean_lj* audio and its pitch-modified versions (0.75× and 1.5×) using PSOLA.

Overall, these results show that PSOLA can modify pitch and duration independently of each other.

1.2 PSOLA vs. resampling

Besides PSOLA, we also experimented with a naive pitch modification method based on resampling. To modify the pitch by factors of 0.75 and 1.5, we resample the audio waveforms using resampling ratios of $\frac{1}{0.75}$ and $\frac{1}{1.5}$, respectively, while maintaining the original sampling rate.

Table 2 compares the effects of PSOLA and naive resampling on pitch, formant frequencies, and duration. Both methods achieve F0 ratios close to the target pitch factors of 0.75 and 1.5. However, PSOLA preserves the original duration while resampling changes the duration to approximately 1.333 and 0.667, respectively.

Furthermore, PSOLA generally preserves the original formant frequencies, with F1 and F2 ratios close to 1.0, while resampling changes both formant frequencies, which can make listeners think they are hearing a different speaker.

Table 2: Comparison of PSOLA and resampling for pitch modification. Ratio means the value after a change divided by the value before it, and F0, F1 and F2 mean the medians over voiced frames.

Clip	Modification	Duration Ratio	F0 Ratio	F1 Ratio	F2 Ratio
clean_lj	PSOLA ×0.75	1.000	0.752	1.035	1.017
	Resample ×0.75	1.333	0.754	0.831	0.953
	PSOLA ×1.5	1.000	1.495	1.002	0.984
	Resample ×1.5	0.667	1.512	1.405	0.847
clean_jfk	PSOLA ×0.75	1.000	0.775	1.020	0.995
	Resample ×0.75	1.333	0.757	0.835	0.893
	PSOLA ×1.5	1.000	1.496	0.997	0.996
	Resample ×1.5	0.667	1.522	1.429	1.173

Figure 2 compares the spectrograms of the first 3 seconds of *clean_lj* for the original signal, the PSOLA ×1.5 version, and the resampled ×1.5 version. It shows that PSOLA largely preserves the original formant structure while increasing the pitch (i.e., increasing the spacing between harmonics).

In contrast, resampling compresses the audio in time and shifts the formant frequencies, resulting in a different spectral structure. This figure further shows that PSOLA can modify pitch while preserving the original duration and formant structure, whereas resampling changes all three.

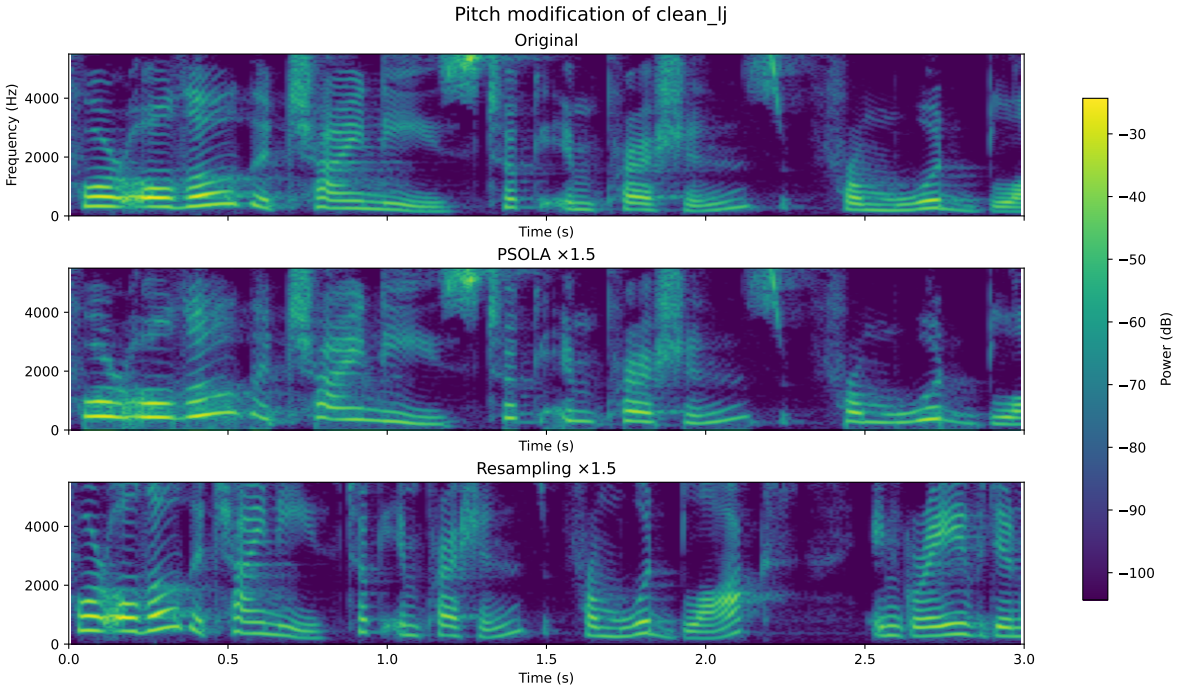


Figure 2: Spectrograms of the first 3 seconds of *clean_lj* for the original audio, the PSOLA ×1.5 version, and the resampled ×1.5 version.

1.3 Your own voice

In this section, we apply audio modification methods using both PSOLA and resampling to our own recordings. We perform the following modifications:

- PSOLA with the pitch × 0.75
- PSOLA with the pitch × 1.5
- a PSOLA monotone, and
- a resampled version with the pitch × 1.5

For audio analysis, we use a pitch range of 50–300 Hz and a formant ceiling of 5000 Hz. For the original recording, the estimated F0 is 96.955 Hz, with a standard deviation of 31.211 Hz. The estimated F1 and F2 are 487.395 Hz and 1672.228 Hz, respectively, and the total duration is 18.155 seconds.

Similar to the previous section, we also calculate the duration, F0, F1, and F2 ratios, along with the F0 standard deviation, for each modified recording, as shown in Table 3. Unlike resampling, which changes all four measurements, PSOLA preserves the original duration and largely maintains the F1 and F2 ratios while modifying F0.

Among all the generated samples, the PSOLA ×0.75 version sounds the most similar to my original voice. Since my voice is naturally low-pitched, I am more familiar with hearing myself speak at a lower pitch. Although the PSOLA ×1.5 version still sounds like me, its pitch is higher than what I would

normally produce. The monotone version sounds very robotic, which is consistent with its low F0 variation. Finally, the resampled version sounds the most artificial, as it is sped up and has a higher pitch, making my voice sound like a chipmunk.

Table 3: Measured duration, F0, F1, and F2 ratios and F0 standard deviations for the modified recordings.

Modification	**Duration Ratio**	**F0 Ratio**	**F1 Ratio**	**F2 Ratio**	**F0 Std. (Hz)**
PSOLA ×0.75	1.000	0.761	0.986	0.997	29.806
PSOLA ×1.5	1.000	1.575	0.950	0.953	47.593
PSOLA Monotone	1.000	1.000	0.941	0.963	4.641
Resample ×1.5	0.667	1.527	1.440	1.194	47.486

2 Forced alignment with the Montreal Forced Aligner

2.1 Align and inspect

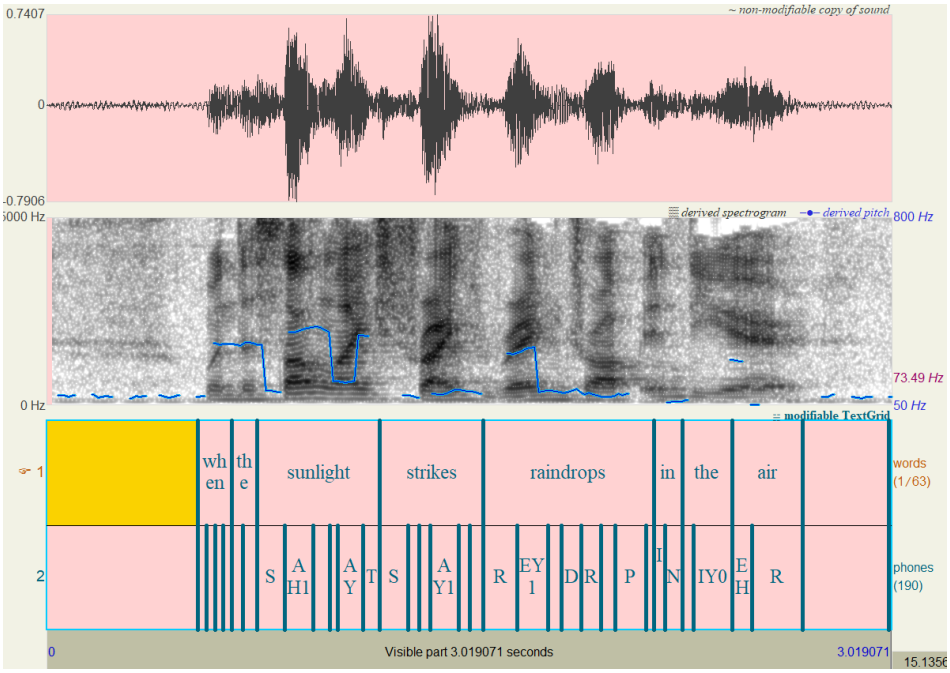
In this section, we use the Montreal Forced Aligner (MFA) with the pretrained English acoustic model (`english_us_arpa`) to align the five recordings with their corresponding transcripts. Table 4 shows the number of words and phones, the start time of the first word, the end time of the last word, and the number of unknown words for each recording. All five recordings have zero unknown words, indicating that all words are covered by the pronunciation dictionary.

We also compare the word start times between *my_voice* and *my_voice_noisy*. The mean difference is 1971 ms, while the maximum difference is 2550 ms, occurring at the word "ends". However, since these two recordings were recorded separately, their word start times have different initial offsets. Specifically, the first word starts at 0.54 seconds in the clean recording and 1.58 seconds in the noisy recording, resulting in an initial offset of 1.04 seconds. Additionally, the noisy recording has a longer speaking duration, indicating differences in speaking rate. Therefore, the differences in word boundaries may be caused by the initial offset and natural variations in speaking rate rather than the background noise itself.

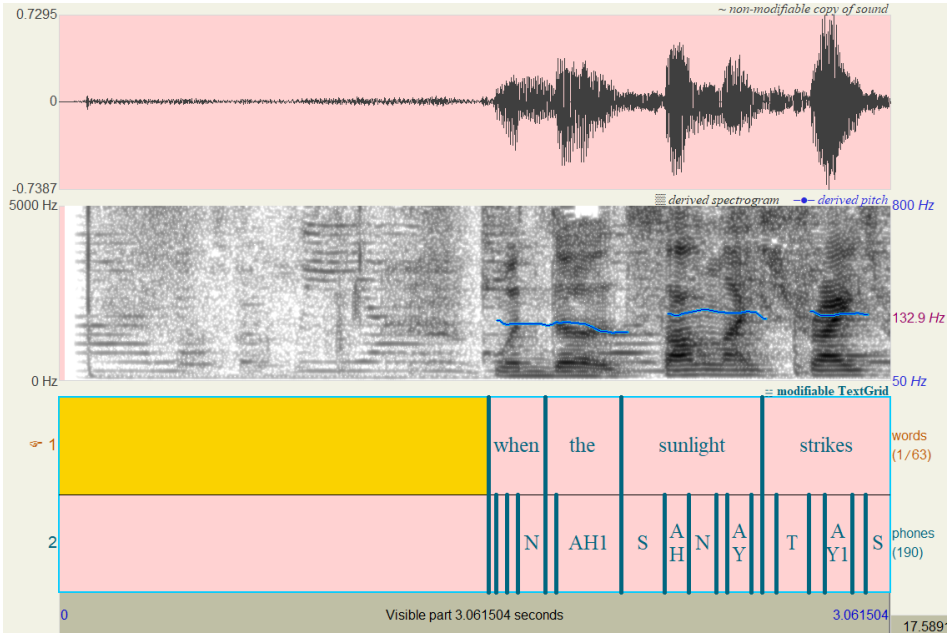
Table 4: MFA alignment statistics for the five recordings, including the number of words and phones, first word start time, last word end time, and number of unknown words.

Utterance	**Words**	**Phones**	**Start (s)**	**End (s)**	**Unknown**
clean_jfk	22	73	0.32	10.48	0
clean_lj	24	104	0.00	9.57	0
my_voice	51	178	0.54	16.81	0
my_voice_noisy	51	178	1.58	19.23	0
my_voice_slower	51	178	0.83	25.20	0

Figure 3 also shows that the two recordings start at different times, while the word and phone boundaries accurately reflect where the corresponding words and phones are spoken. MFA does not appear to be significantly affected by the background music at the beginning, likely because the music is relatively quiet and not included in the transcript.



(a) Clean recording (*my_voice*).



(b) Noisy recording (*my_voice_noisy*).

Figure 3: Screenshots of the clean and noisy recordings in Praat, showing the spectrograms and MFA word and phone alignments.

2.2 Speaking rate and vowels

In this section, we use the obtained word and phone boundaries to analyze the speaking rate and vowel formants of three recordings (*my_voice*, *clean_lj*, and *clean_jfk*). For speaking rate, we report the number of phones per second (pps) between the start of the first word and the end of the last word, the fraction of that time spent in pauses, and the mean phone duration, see Table 5. My recording and *clean_lj* have similar speaking rates, with comparable phones per second and pause fractions. In contrast, *clean_jfk* has a higher pause fraction and a slower speaking rate, which is also noticeable when listening to the audio, as it is delivered in a more formal public-speaking style.

Table 5: Speaking rate stats for the three recordings, including phones per second, pause fraction, and mean phone duration.

Recording	**Phones/sec**	**Pause Fraction**	**Mean Phone Duration (ms)**
my_voice	10.940	0.119	80.5
clean_lj	10.867	0.079	84.7
clean_jfk	7.185	0.308	96.3

For vowel formant analysis, we measure F1 and F2 using the Burg method, taking the median values over the middle half of each stressed vowel based on the aligned phone boundaries. Table 6 shows the median F1, F2, and duration for each vowel that occurs at least three times for each speaker. Only three stressed vowel categories overlap between any two speakers (AE1, AH1, and EH1), making direct comparisons limited. Comparing my voice with *clean_jfk*, my recording generally has lower F1 and F2 values, except for the F1 of AE1. This may suggests a relatively higher and more backed tongue position compared to JFK. A similar trend is observed when comparing with *clean_lj*. However, the limited number of overlapping stressed vowel samples makes these conclusions less reliable, as the medians can be noisy when calculated from fewer than five samples.

Table 6: Median F1, F2, and duration of stressed vowels occurring at least three times for each speaker. The number of vowel tokens is shown in parentheses next to F1.

	F1 (Hz)			**F2 (Hz)**			**Duration (ms)**		
Phone	**my_voice**	**clean_lj**	**clean_jfk**	**my_voice**	**clean_lj**	**clean_jfk**	**my_voice**	**clean_lj**	**clean_jfk**
AE1	835.5 (3)	–	697.1 (4)	1624.5	–	1811.3	160.0	–	225.0
AH1	620.9 (4)	–	756.4 (4)	1076.4	–	1394.8	110.0	–	75.0
AO1	–	–	571.0 (3)	–	–	1215.8	–	–	90.0
AY1	608.9 (5)	–	–	1636.4	–	–	100.0	–	–
EH1	476.7 (6)	690.1 (3)	–	1693.7	1895.3	–	65.0	90.0	–
EY1	474.6 (6)	–	–	1988.3	–	–	100.0	–	–
IH1	440.0 (3)	–	–	1550.8	–	–	60.0	–	–
UH1	–	537.4 (3)	–	–	1364.9	–	–	60.0	–
UW1	–	–	489.6 (4)	–	–	1090.9	–	–	235.0

We also plot the median F1 and F2 values of these vowels in Figure 4, with both axes inverted. Due to the limited number of overlapping stressed vowels, it is difficult to draw conclusions about vocal tract differences between the speakers.

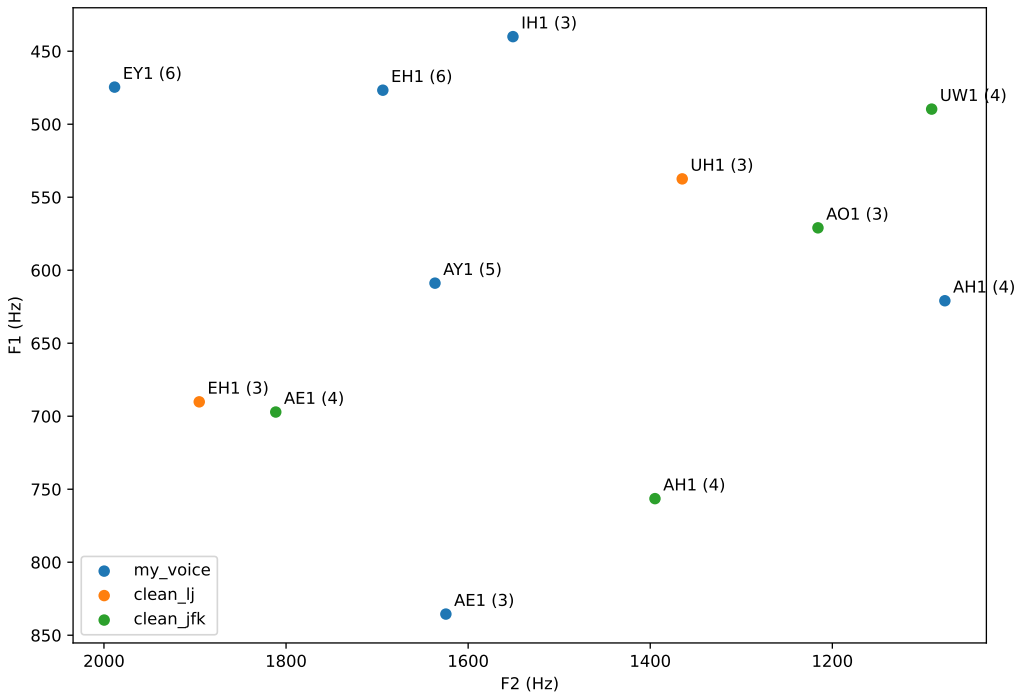


Figure 4: Median F1 and F2 of stressed vowels for the three speakers, with both axes inverted.

2.3 Alignment of slowed speech

In this section, we perform MFA on the same *my_voice* recording after stretching it by a factor of 1.5 using PSOLA. Ideally, the word and phone durations should also increase by a factor of 1.5.

From the experiment, the median word duration ratio (new duration divided by original duration) is 1.476, with a minimum of 1.333 and a maximum of 2.000. The median duration ratio is close to the expected value of 1.5, suggesting that MFA can accurately locate word boundaries even after the duration modification using PSOLA.

The four most extreme words are two occurrences of "the", both with a ratio of 1.333, and two occurrences of "a", with ratios of 1.833 and 2.000, respectively. These words that deviate significantly from the expected ratio are all short words, so the deviations are more likely caused by alignment errors than by PSOLA itself, since small differences in the detected word boundaries can lead to large duration ratios for short words.

3 Text-to-speech and speech recognition

3.1 Text to speech

In this section, we use the `facebook/mms-tts-eng` model to synthesize audio from the given sentences. Table 7 shows the audio duration, synthesis time, real-time factor (RTF), and tokenized text for each sentence. All sentences have an RTF below 1.0, indicating that the model can synthesize speech faster than real time.

Among all the synthesized samples, sentences containing numbers are all not pronounced correctly. For example, the model fails to correctly pronounce "9:30" in sentence 4, "25 dollars" in sentence 5, and the phone number "555-0142" in sentence 6. In sentence 4, the tokenizer removes "9:" and only keeps "30", which explains why the generated speech is incorrect.

The abbreviation category also has pronunciation issues. For example, "Dr." in sentence 7 should be pronounced as "doctor", but the model fails to produce the correct pronunciation. Similarly, for sentence 8, which contains "NASA" and "FBI", the tokenizer converts them to lowercase without expanding the abbreviations. As a result, the model does not pronounce "FBI" as the individual letters "F", "B", and "I", but instead attempts to pronounce it as a word.

Table 7: TTS synthesis results for 12 sentences, including the text retained by the tokenizer, audio duration, synthesis time, and real-time factor (RTF).

ID	Tokenized Text	Duration (s)	Time (s)	RTF
1	the quick brown fox jumps over the lazy dog	3.904	0.692	0.177
2	she sells seashells by the seashore	2.640	0.398	0.151
3	please call me when you get home tonight	2.928	0.419	0.143
4	the meeting starts at 30 on march 5th	2.784	0.401	0.144
5	i paid 25 dollars for 3 tickets	2.768	0.449	0.162
6	call 555-0142 before noon	2.448	0.376	0.154
7	dr smith lives on elm st in houston	4.880	0.655	0.134
8	nasa and the fbi shared the data	2.672	0.434	0.162
9	siobhan met nguyen at the worcester station	3.504	0.483	0.138
10	i will read the book you read yesterday	3.504	0.456	0.130
11	the wind will wind around the old road	2.704	0.410	0.152
12	can you tell me how far it is from the library to the train station and whether the bus runs on sundays	6.480	0.878	0.136

3.2 Speech recognition

In this section, we use both `whisper-tiny` and `whisper-base` with greedy decoding to transcribe the synthesized samples and compute the WER of every transcript against its reference. When computing WER, we convert all text to lowercase and remove punctuation to ensure that the scores are not affected by differences in capitalization or punctuation. Table 8 shows the mean WER of synthesized speech across different sentence types. Sentences containing abbreviation (i.e. "Dr. Smith lives on Elm St. in Houston.") have the highest WER (0.681), followed by numbers (0.594) and name (0.571).

Table 8: Mean WER of synthesized speech across different sentence types.

Sentence Type	Mean WER
Abbreviation	0.681
Numbers	0.594
Name	0.571
Plain	0.287
Long	0.114
Homograph	0.000

Table 9 compares the performance of `whisper-tiny` and `whisper-base` on synthesized, real clean, and real noisy speech. Both models achieve lower WER on real speech than on synthesized speech, which is consistent with the pronunciation errors we observed and discussed in the previous section. `whisper-base` performs slightly better compares to `whisper-tiny` on all three types of speech.

Table 9: Mean WER of Whisper-tiny and Whisper-base on synthesized, real clean, and real noisy speech.

Model	Synthesized	Real Clean	Real Noisy
Whisper-tiny	0.422	0.087	0.137
Whisper-base	0.360	0.087	0.101

However, for `whisper-base`, sentence 2, which is the plain sentence type, has the highest reported WER of 0.833. The reference and recognized transcriptions are:

- **Reference:** she sells seashells by the seashore
- **Recognized:** she sells sea shells but the sea shore

The high WER is mainly caused by word segmentation errors, where "seashells" and "seashore" are recognized as two separate words, "sea shells" and "sea shore", respectively. Additionally, "by" is incorrectly recognized as "but".

There are two main types of errors that contribute to high WER. The first is actual recognition errors made by the ASR model. This can be observed from the difference between real clean and real noisy speech, where background noise makes it harder for ASR models to accurately recognize the spoken content. Additionally, uncommon names such as "Siobhan" may also be incorrectly transcribed.

The second type of error is caused by the lack of text normalization before feeding the input into the TTS model. Sentences containing abbreviations (e.g., Dr.", St.", and FBI") or numbers (e.g., 9:30" and "555-0142") are not synthesized correctly because these expressions are not properly converted into their spoken forms. As a result, the generated speech differs from the intended pronunciation, leading to higher WER.

3.3 Does ASR survive PSOLA?

In this section, we apply PSOLA and resampling modifications to three recordings (*clean_lj*, *clean_jfk*, and *clean_libri*). For pitch modification, we use factors of 0.6, 0.8, 1.0, 1.25, 1.6, and 2.0 for both PSOLA and resampling. For duration modification, we use PSOLA with factors of 0.5, 0.75, 1.0, 1.5, and 2.0. We then transcribe all modified recordings using the `whisper-base` model.

Figure 5 shows the mean WER across the three recordings for different pitch and duration factors, with the PSOLA and resampling pitch curves plotted on the same axis. The results show that PSOLA pitch modification maintains a relatively stable WER across all factors, while resampling leads to a significant increase in WER at more extreme pitch factors, particularly 0.6 and 2.0. This is consistent with our observations in Section 1.2, where resampling alters the spectral structure of the audio, making the speech more difficult for the ASR model to recognize. For duration modification, the WER remains relatively stable, with a slight increase when the audio is significantly shortened or lengthened.

3.4 Your own voice

Finally, in this section, we apply ASR to our own recording, a noisy version with added background music, and a TTS-generated version of the same script. Table 10 shows the WER against the ground-truth script for all three types of speech using both models.

One interesting finding is that the noisy version achieves a lower WER than the clean version for both models, while the clean and TTS versions have relatively similar WER. One possible explanation is that the added noise is only background music and is not loud enough to significantly affect ASR performance.

To further investigate these results, we compare the predicted transcripts with the ground-truth transcript. The normalized text is shown below:

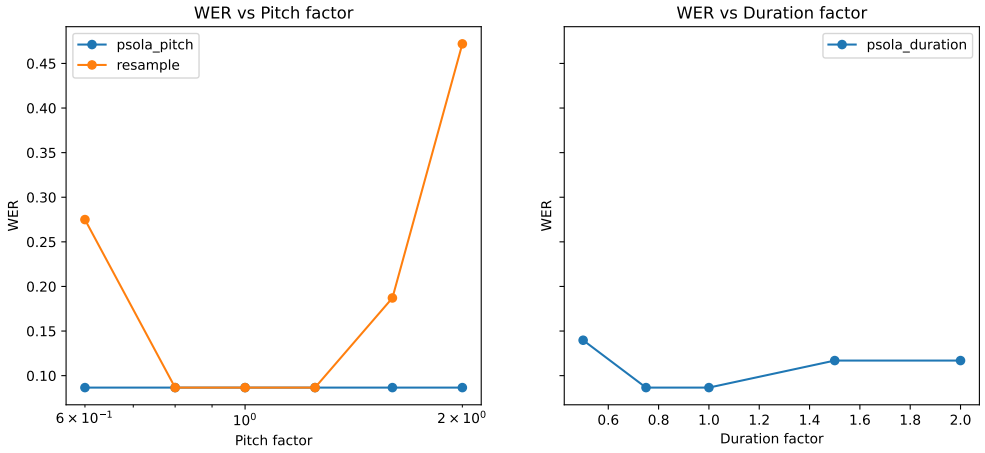


Figure 5: Mean WER of `whisper-base` across three recordings under different pitch and duration modification factors. The left plot compares PSOLA and resampling for pitch modification (plotted pitch factors on log scale), while the right plot shows PSOLA duration modification.

Table 10: WER of `whisper-tiny` and `whisper-base` on our own clean, noisy, and TTS-generated recordings.

Model	**Clean**	**Noisy**	**TTS**
Whisper-tiny	0.216	0.118	0.176
Whisper-base	0.157	0.078	0.157

- GT: when the sunlight strikes raindrops in the air they act as a prism and form a rainbow the rainbow is a division of white light into many beautiful colors these take the shape of a long round arch with its path high above and its two ends apparently beyond the horizon
- my_voice (clean): when the sunlight strikes rain drop in the air they access the prism and form a rainbow the rainbow is a division of white light into many beautiful colors these take the shape of a longround art with its path high above and its two ends apparently beyond the horizon
- my_voice_noisy: with the sunlight strikes raindrops in the air they act as a prison and form a rainbow the rainbow is a division of white light into many beautiful colors these take the shape of a long round art with its path height above and its two ends apparently beyond the horizon
- tts: when the sun likes dry rain drops in the air they act as a prism and form a rainbow the rainbow is a division of white light into many beautiful collars these take the shape of a long round arc with its path high above and its too ends apparently beyond the horizon

We observe several word segmentation errors in the clean recording, such as "raindrops" being transcribed as "rain drop" and "long round" as "longround". Additionally, some parts of my recording are spoken too quickly, which may make the speech harder for the ASR model to recognize. For example, "act as a" is incorrectly transcribed as "access the". In the noisy version, we also observe recognition errors such as "prism" being transcribed as "prison" and "arch as "art".

Overall, these results show that WER can be affected by word segmentation and pronunciation differences, even when the spoken content is mostly understandable. Adding background noise does not significantly hurt ASR performance, possibly because the added noise is relatively weak.

Code

The code used to reproduce the experiments, results, and figures in this report is available at:

`github.com/Morris88826/CSCE630_Speech_Processing/tree/main/hw2`